

1. Administrative & Study Identification

Full Study Title: ELEMENT-MDS: A Study to Compare the Efficacy and Safety of Luspatercept in Participants With Myelodysplastic Syndrome (MDS) and Anemia Not Receiving Blood Transfusions

Official Title: A Phase 3, Open-label, Randomized Study to Compare the Efficacy and Safety of Luspatercept (ACE-536) vs Epoetin Alfa for the Treatment of Anemia Due to Revised International Prognostic Scoring System (IPSS-R) Very Low, Low, or Intermediate-Risk Myelodysplastic Syndrome (MDS) in Erythropoiesis-Stimulating Agent (ESA)-Naïve Participants Who Are Non-Transfusion Dependent (NTD): The "ELEMENT-MDS" Trial **Short Title/Acronym:** ELEMENT-MDS **Protocol Number:** CA056-025 **NCT Number:** NCT05949684 **Trial Status:** RECRUITING **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Luspatercept (BMS-986346 / ACE-536 / Reblozyl®) vs. Epoetin Alfa (Procrit, Epogen, Eprex, Retacrit) **Phase of Development:** Phase 3 **Medical Monitor:** Bristol Myers Squibb Clinical Trials Contact

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the efficacy and safety of luspatercept versus epoetin alfa in reducing progression to red blood cell (RBC) transfusion dependence in ESA-naïve, non-transfusion dependent (NTD) adult patients with anemia due to lower-risk MDS (LR-MDS). **Secondary Objectives:** To evaluate hemoglobin (Hb) increase ≥ 1.5 g/dL sustained for continuous 16-week and 24-week intervals, time to RBC transfusion dependence, transfusion-free survival, time to achieve modified hematologic improvement-erythroid response, extended transfusion independence, and quality of life (QoL).

2.2 Background & Rationale Anemia is a common, debilitating symptom in patients with myelodysplastic syndromes (MDS), frequently leading to dependence on RBC transfusions. Transfusion-dependent patients face worse overall survival and quality of life, alongside complications like iron overload. While ESAs are standard for lower-risk MDS, luspatercept (an erythroid maturation agent) has proven effective in treating transfusion-dependent patients. This study addresses the medical necessity of evaluating luspatercept earlier in the treatment paradigm to prevent or delay RBC transfusion dependence in the majority NTD LR-MDS patient population who are ESA-naïve.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Epoetin Alfa) **Blinding:** Open-label
Randomization: Randomized (1:1 parallel assignment) **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration Target \$\$\$: 360 patients **Number of Sites:** Global (United States, Australia, Brazil, Canada, Chile, China, etc.) **Estimated Study Start:** July 2023 **Estimated Primary Completion:** 96-week treatment period

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Documented diagnosis of MDS according to World Health Organization (WHO) 2016 that meet IPSS-R classification of very low, low, or intermediate-risk disease (intermediate-risk of ≤ 3.5 IPSS-R score) confirmed via bone marrow aspirate and: i) $< 5\%$ blasts in bone marrow and $< 1\%$ blasts in peripheral blood. 2. Participant is not transfusion dependent (NTD) based on IWG2018 criteria. 3. Participant is erythropoiesis-stimulating agent (ESA) naive. Participants may be randomized at the investigator's discretion if the participant received no more than 2 prior doses of epoetin alfa, epoetin alfa biosimilar, or darbepoetin alfa, with the last dose at least 8 weeks prior to randomization. 4. Participant has a baseline endogenous serum erythropoietin (sEPO) level of ≤ 500 U/L. 5. Participant has symptoms of anemia: i) Participant records a severity score of "moderate" or greater on at least 1 PGI-S item of fatigue, weakness, shortness of breath, or dizziness performed during the screening period. 6. Participant has a baseline Hb concentration prior to randomization of ≤ 9.5 g/dL. The baseline Hb will be calculated using the mean of the two lowest available Hb measurements within 16 weeks prior to randomization and must include at least one central lab Hb reading done within the screening period (no more than 35 days before randomization). The two Hb measurements must have been performed at least seven days apart. Hb levels less than 21 days following RBC transfusion should not be used. Split samples for local assessments are not required.

4.2 Exclusion Criteria 1. Participant with secondary MDS (that is, MDS that is known to have arisen as the result of chemical injury or treatment with chemotherapy and/or radiation for other diseases). 2. Participant with known history of diagnosis of AML. 3. Participant with a history of cerebrovascular accident (including ischemic, embolic, and hemorrhagic cerebrovascular accident), transient ischemic attack, deep venous thrombosis (including proximal and distal), pulmonary or arterial embolism, arterial thrombosis, or other venous thrombosis within 6 months prior to randomization. 4. Participant with a history of pure red

cell aplasia and/or antibody against erythropoietin. 5. Other protocol-defined Inclusion/Exclusion criteria apply.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Luspatercept:** Starting dose of 1.0 mg/kg (escalation up to 1.75 mg/kg allowed) once every 3 weeks. * **Epoetin alfa:** Starting dose of 450 IU/kg (escalation up to 1050 IU/kg allowed) once a week. **Route of Administration:** Subcutaneous (Biological injections) **Duration of Treatment:** 96 weeks

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for MDS-associated anemia. **Prohibited:** More than 2 prior doses of ESA (epoetin alfa, epoetin alfa biosimilar, or darbepoetin alfa), or receipt of an ESA within 8 weeks prior to randomization.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Up to 5 weeks (35 days)	Informed Consent, MDS Diagnosis Confirmation via bone marrow aspirate, IPSS-R score, sEPO level, Hb central lab reading
2	Baseline	Day 0
3	Randomization (1:1), First Dose	Interim
4	Weeks 1-96	Treatment administration, Efficacy Assessments (Hb measurements, RBC transfusion tracking), Safety Labs
5	Week 96	Primary Completion Evaluation, Transition to extension phase / post-treatment follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Number of participants with lower-risk non-transfusion dependent myelodysplastic syndromes (NTD-MDS) who converted to Transfusion Dependence (TD) during any continuous 16-week interval within the 96-week treatment period. TD is defined as ≥ 3 red blood cells (RBC) units/16 weeks assessed by International Working Group (IWG) 2018.

7.2 Secondary & Exploratory Endpoints * Number of participants with an increase from baseline in mean Hb values of ≥ 1.5 grams/deciliter (g/dL) in any continuous 16-week interval within the 48-week Treatment Period in the absence of transfusion. * Number of participants with an increase from baseline in mean Hb values of ≥ 1.5 g/dL in any continuous 24-week interval within the 48-week and 96-week treatment periods in the absence of transfusion. * Time to RBC transfusion dependence and transfusion-free survival. * Time to achieve modified hematologic improvement-erythroid response.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of type, severity, and relationship to treatment of adverse events throughout the trial. **Serious Adverse Events (SAEs):** Evaluated for occurrences of progression to high-risk MDS and acute myeloid leukemia (AML), as well as overall survival. **Data Monitoring Committee (DMC):** Standard internal/external safety monitoring mechanisms maintained by Bristol Myers Squibb.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy assessments. **Sample Size Justification:** Target enrollment of $N=360$ powered to compare the efficacy of luspatercept vs epoetin alfa in delaying/preventing TD progression. **Handling of Missing Data:** Standard clinical trial imputation strategies (e.g., treating missing transfusion evaluation periods as non-responders where applicable).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring across global sites. **Audit Trail:** Utilizing validated eCRF tracking to consistently record Hb values (mean of two lowest readings prior to baseline performed at least 7 days apart) and central laboratory validation for initial sEPO limits.

11. Study Identifiers & Governance

Primary ID: CA056-025 **Registry Number:** NCT05949684 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** ELEMENT-MDS: A Study to Compare the Efficacy and Safety of Luspatercept in Participants With Myelodysplastic Syndrome (MDS) and Anemia Not Receiving Blood Transfusions **Official Regulatory Title:** A Phase 3, Open-label, Randomized Study to Compare the Efficacy and Safety of Luspatercept (ACE-536) vs Epoetin Alfa for the Treatment of Anemia Due to Revised International Prognostic Scoring System (IPSS-R) Very Low, Low, or Intermediate-Risk Myelodysplastic Syndrome (MDS) in Erythropoiesis-Stimulating Agent (ESA)-Naïve Participants Who Are Non-Transfusion Dependent (NTD): The "ELEMENT-MDS" Trial

12. Clinical Framework (MDS Specific)

Primary Condition: Myelodysplastic Syndromes (MDS) / Anemia
Diagnosis Threshold: IPSS-R Very Low, Low, or Intermediate-Risk (≤ 3.5), Baseline sEPO ≤ 500 U/L, Baseline Hb ≤ 9.5 g/dL
Medical Methodology: Erythroid maturation agent (Luspatercept) vs Erythropoiesis-stimulating agent (Epoetin alfa)
Optimization Target: Prevention of progression to Red Blood Cell (RBC) Transfusion Dependence

13. Study Procedures & Methodology

Management Pathway: Active intervention addressing moderately severe chronic anemia in NTD LR-MDS
Education Protocol (HCP): Administration of respective biologic injections (subcutaneous dosing specific to Luspatercept vs Epoetin Alfa frequencies)
Education Protocol (Patient): Patient Global Impression of Severity (PGI-S) self-reporting for anemia symptoms (fatigue, weakness, shortness of breath, dizziness)
Quality Audit Mechanism: Continuous Hb measurement tracking and stringent validation of transfusion volumes (≥ 3 units per 16 weeks mapping)

14. Observation & Follow-Up Schedule

Total Duration: 96 Treatment weeks + Extension phase follow-up
Onsite Visit Cadence: Every 1 to 3 weeks (aligned to match epoetin alfa weekly or luspatercept tri-weekly injection schedules)
Remote Monitoring Frequency: Standard safety check intervals between doses.
Checklist Review Interval: Evaluated systematically in continuous 16-week rolling intervals to capture TD thresholds.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				